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MILITARY SPECIFICATION

RADIAC CALIBRATOR AN/UDM-2( )

This specification is approved for use by Electronics Research and Development Command, Department of the Army, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for Radiac Calibrator which provides the facilities for checking the operational reliability and calibration accuracy of various radiacmeters and radiac set. The calibrator consists of a dosimeter jig assembly and a dose rate jig assembly; each assembly can be utilized independently of the other (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The effective issue or revision of the following documents shall be that listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, unless (i) specific issues are set forth therefor in the cited specifications, or (ii) issues different than those specified in the cited specifications are set forth in the solicitation. The date of the applicable DODISS and supplements thereto shall be as specified in the solicitation or contract.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation Packing, Methods of
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	- Marking of Electronic Items
MIL-F-14072	- Finishes for Ground Electronic Equipment

Beneficial comments (recommendations, additions, deletions and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, Combat Surveillance and Target Acquisition Laboratory, ATTN: DELCS-PE, Fort Monmouth, NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.)

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-810	- Environmental Test Methods

DRAWINGS

DL-SM-B-508965	- Radiac Calibrator AN/UDM-2( )
SPI 1G00152	- Special Packaging Instruction

OTHER PUBLICATIONS

US Nuclear Regulatory Commission Regulations - Title 10, Chapter I, Code of Federal Regulations, Parts 19, 20, 21, 30, 51, 71 and 110.

US Department of Transportation Regulations - Title 49, Code of Federal Regulations

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Construction. The equipment shall be constructed in accordance with the requirements of this specification, and of DL-SM-B-508965.

3.2 First article. When specified in the contract or purchase order, the contractor shall furnish first article units as required and in the quantities specified therein (see 4.3).

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply (see 4.4).

3.3.1 Finish. All surfaces of the AN/UDM-2( ) requiring a protective coating shall be finished in accordance with MIL-F-14072 (see 4.4).

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3.4 Performance.

3.4.1 Calibration accuracy.

3.4.1.1 Dosimeter jig assembly (lower source). The dose rate produced by the lower sources shall be within ± 5R/min when related to a secondary calibration standard. When measured in accordance with 4.6.1.1, using a government furnished Victoreen Model 555 Radacon II, the dose rate produced by the lower sources shall be 205 R/min ± 5R/min.

3.4.1.2 Dosimeter jig assembly (upper source). The dose rate produced by the upper source shall be within ± 0.05 mR/min when related to a secondary calibration standard. When measured in accordance with 4.6.1.2, using a government furnished Victoreen Model 555 Radacon II, the dose rate produced by the upper source shall be 0.95 mR/min ± 0.05 mR/min.

3.4.1.3 Dose rate jig assembly. All measured test position dose rate shall be within ± .5R/min when related to a secondary calibration standard. When measured in accordance with 4.6.1.3, using a government furnished Victoreen Model 555 Radacon II, the dose rate produced with the shutter in the 100 R/hr position shall be 44.5 R/min ± 0.5 R/min.

3.4.1.4 Radioactivity. The government shall be provided with the maximum quantity of radioactive material contained in each sealed source needed to produce the required radiation output for compliance with 3.4.1.1, 3.4.1.2 and 3.4.1.3. Each sealed source shall not exceed the maximum quantity of radioactive material specified and shall in accordance with:

a. Special form material test requirements of Title 49, Code of Federal Regulations of US Department of Transportation regulations.

b. American National Standards Institute N542-1977 classification of at least 43343.

c. Approval by the US Nuclear Regulatory Commision or Agreement State for its integrity of construction.

The government shall be provided with supporting documentation indicating compliance with a, b, and c above.

3.4.2 Compatibility. Radiacmeters IM-9( )/PD, IM-93( )/UD and IM-147( )/PD, when exposed in the dosimeter jig assembly for the time specified shall respond in accordance with the appropriate reading and tolerance specified in figure 3-1 (see 4.7.1). Radiacmeters IM-174A/PD and IM-174B/PD and Radiac Set AN/PDR-27( ), when calibrated in the dose rate jig assembly in accordance with the instructions shall respond in accordance with the appropriate reading and tolerance specified in figure 3-2 (see 4.7.2).

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FIGURE 3-1. Dosimeter discharge data.

I Dosimeter	II Discharge position	III Manufacturer	IV Procurement order RR	V Year	VI Time	VII Discharge reading ± 20%	VIII Discharge reading limit ±20 %
IM-93	lower	Landsverk	21505-P	56	5 min	300	240-360
0-600 R	lower	Bendix	30884-PP	57	5 min	405	324-486
	lower	Bendix	4371-PP	60	5 min	70	56-84
	lower	Bendix	15916-PP	63	5 min	60	48-72
	lower	Landsverk	4596-PP	61	5 min	40	32-48
	lower	Landsverk	15631-PP	62	5 min	40	32-48
	lower	Landsverk	DAAB05-68-C-0911	68	5 min	300	240-360
IM-147	lower	Bendix	3439-PP	59	1 min-45 sec	25	20-30
0-50	lower	Landsverk	DAAB05-68-C-0911	68	40 sec	40	32-48
IM-9E	upper	Bendix	15895-PP	63	2 min	100	80-120
0-200 mR	upper	Landsverk	15580-PP	62	3 min	100	80-120
	upper	Landsverk	DAAB05-68-C-1678	67	2 min	80	64-96
IM-9F	upper	Landsverk	C2537MFR-00910	68	5 min	80	64-96
CDV-138	upper	Bendix	No indication		2 min	100	80-120
0200 mR							
CDV-742	lower	Bendix	No indication		2 min	120	96-144
0200 R	lower	Landsverk	No indication		5 min	120	96-144

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FIGURE 3 - 2. Dose rate jig assembly.

1. IM-174( )/PD; IM-174A/PD; IM-174B/PD

   a. Release shipping lock
   b. Remove detector assembly from IM-174( )/PD. Position it securely in drawer. Switch on IM-174( )/PD. Set it up per its instruction plate.
   c. Close drawer
   d. Rotate shutter to 100 R/hr position. Note meter reading.
   e. Rotate shutter to 10 R/hr position. Note meter reading.
   f. Meter should read:
      (1) IM-174 between 177 and 117 at 100 R/hr and between 18 and 11 at 10 R/hr.
      (2) IM-174(A) and IM-174(B) between 120 and 80 at 100 R/hr and between 12 and 8 at 10 R/hr.

      If not, remove calibration control cover. Adjust IM-174( )/PD until it reads in tolerance at both positions.
   g. Close shutter. Open drawer. Remove detector assembly. Replace it in the IM-174( )/PD.
   h. Close drawer locate and close shipping lock. Lock source shutter with the key.

2. AN/PDR-27 J, L, R and S

   a. Release shipping lock.
   b. By removing bolts from probe clips, separate both probe units.
   c. Check that spacer block is completely inside drawer.
   d. Close the drawer. Insert smaller probe into hole in handle end of drawer. Switch to 500 mR/hr range and rotate shutter to 100 R/hr position. Rotate small probe to achieve minimum and maximum readings. Note readings.
   e. The average reading shall be 200 mR/hr $\pm$ 30% or 140 to 260 mR/hr.
   f. Rotate shutter to off position. Remove probe unit and open drawer. Remove spacer block and turn over to fit on handle close drawer.
   g. Insert smaller probe in new position. Switch to 50 mR/hr range. Rotate shutter to 100 R/hr position. Rotate smaller probe to achieve minimum and maximum readings. Note readings.
   h. The average reading shall be 30 mR/hr $\pm$ 30% or 21 to 39 mR/hr.
   i. Rotate shutter to off position. Remove probe from drawer.
   j. Switch to 5 mR/hr position and place large probe on the top rear edge of drawer unit, opened to its fullest extremity. Leave shutter in off position, and observe readings.
   k. Average reading should be 3.2 mR/hr $\pm$ 30% or 2.2 to 4.2 mR/hr.
   l. Place large probe on uper edge of handle side of box. Switch to 0.5 mR/hr position and observe readings.
   m. Average reading should be 0.30 mR/hr $\pm$ 30% or 0.21 to 0.39 mR/hr.
   n. If readings obtained are outside quoted tolerances, remove cover over calibration controls and adjust appropriate control to correct reading.
   o. Screw down shipping lock. Lock source shutter with key.
   p. Reassemble AN/PDR-27 probe assembly.

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3.4.3 Surface dose rate. The maximum allowable dose rate at any point on the external surface of the calibrator set shall not exceed 5 mR/hr. This requirement applies under all conditions of storage (see 4.8).

3.5 Weight. The weight of the equipment, less manual, shall not exceed 40 lbs (18.14 Kg) (see 4.13).

3.6 Service conditions.

3.6.1 General. The equipment shall perform as required by this specification when exposed to any operating condition listed in 3.6.2, or after exposure to any storage or transportation condition listed in 3.6.3, or to any possible combination of these service conditions and shall have no corrosion deterioration, physical degradation, or change in tolerance limits which could affect operational service or maintenance requirements. (see 4.9)

3.6.2 Operating conditions. The equipment shall meet the following requirements:

3.6.2.1 Temperature. The equipment shall operate continuously at any ambient temperature from -25°F to +125°F (-31.67°C to 51.67°C) (see 4.9.1 and 4.9.2).

3.6.2.2 Altitude. The equipment shall meet specification performance of altitudes up to and including 10,000 feet (3048m) above sea level (see 4.9.5.1).

3.6.2.3 Humidity. The equipment shall meet specification performance with up to 100 per cent relative humidity at varying temperatures including conditions wherein condensation takes place on the equipment (see 4.9.6).

3.6.2.4 Salt fog. The equipment shall meet 3.4.2 after being subjected to salt fog such as is encountered in coastal regions or on board ship (see 4.9.9).

3.6.3 Storage and transportation conditions.

3.6.3.1 Temperature. The equipment shall be capable of withstanding, without deterioration damage or degradation of performance, long periods of storage at temperature of -65°F to +155°F (-53.89°C to +68.33°C) (see 4.9.1 and 4.9.2).

3.6.3.2 Altitude. The equipment shall meet 3.4.1 after being subjected to altitudes up to and including 50,000 feet (15,240m) above sea level (see 4.9.5.2).

3.6.3.3 Immersion. The equipment shall show no evidence of leakage after it is tested as specified in 4.9.3.

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3.6.3.4 Bench handling. The equipment shall meet the requirement of 3.4.2 without degradation of performance after being subjected to the test of 4.9.4.

3.6.3.5 Bounce. The equipment shall meet the requirement of 3.4.2 without degradation of performance after being subjected to the test of 4.9.8.

3.6.3.6 Vibration. The equipment shall meet the requirement of 3.4.2 after being subjected to vibration such as encountered in a field service environment (see 4.9.7).

3.6.3.7 Rain. The equipment shall meet the requirement of 3.4.2 without degradation of performance after being subjected to the test of 4.9.12.

3.6.3.8 Shock. The equipment shall meet the requirement of 3.4.2 after being subjected to shock such as is encountered in transportation via aircraft, or dropping by parachute (see 4.9.10).

3.6.3.9 Fungus. The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.9.11. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finished or metal surface.

3.7 Special capability requirements.

3.7.1 Facilities. The contractor is required to possess in his plant, or have direct access to a cobalt 60 source capable of providing a minimum field intensity of 125 R/hr at a distance of 1 meter from its geometrical center.

3.7.2 Equipment. The contractor is required to possess in his plant a Victoreen Condenser "R" meter, or equivalent, calibrated by the National Bureau of Standards or other agency approved by the government for measuring radiation flux.

3.7.3 License. The contractor is required to possess or obtain a byproduct material license for the AN/UDM-2( ) and its sources from the US Nuclear Regulatory Commission as per Title 10, Chapter I, Code of Federal Regulations, Part 30.3, or from an agreement-state in which located as per the agreement-state regulations.

3.8 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.12).

3.9 Marking.

3.9.1 General. Marking shall conform to MIL-M-13231.

3.9.2 Radioactive warnings. In addition to the general markings required by 3.9.1, the contractor shall comply with the requirements of Title 10, Code of Federal Regulations, Part 20. Each source shall be etched or stamped with a unique serial number and some symbol identifying the manufacturer and model

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number of the source. The contractor shall provide the government with a list specifying the serial number of each calibrator and the serial number, symbol and nomenclature of each source contained therein.

3.9.3 Serial numbers. Each equipment shall have a serial number in accordance with the requirements of MIL-M-13231, SM-B-509016 and SM-B-509028.

3.10 Systems safety engineering.

3.10.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance with configuration changes and parts selection with requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection (see 4.11).

3.10.2 Corner and edge rounding. All exposed corners and edges shall be rounded to eliminate possible injury to personnel due to lacerations and cuts.

3.10.3 Radioactive materials. Radioactive materials shall not be used (e.g. luminous dials/markings, electron tubes, surge arrestors and lenses).

3.11 Radiological tests. The equipment shall be subject to the tests of 4.10 in the order given. Sources or calibrators that do not meet the limits specified in the source and contamination tests of 3.11.1 and 3.11.2 are not acceptable.

3.11.1 Source tests. Each sealed source shall be subjected to the tests of 4.10.1.1, 4.10.1.2, 4.10.1.3, 4.10.1.4 and 4.10.1.5 in that order.

3.11.1.1 Source leak tests. This test is to be performed after the first encapsulation and again after the second encapsulation. During tests outlined in 4.10.1.1 and 4.10.1.3, any indication of bubbles from the source or on the source indicates the source may be leaking and is not acceptable for use in a calibrator.

3.11.1.2 Source bloat tests. Any indication of bloating (bulging) of the thin window in excess of 0.030 inch from internal pressure at the end of tests 4.10.1.2 and 4.10.1.4 will be considered failure of the test.

3.11.1.3 Source contamination test. Each sealed source to be used in a radiac calibrator shall be subjected to the tests of 4.10.1.5 before it is installed in the calibrator. The results of these tests shall not exceed 0.001 microcuries of removable radioactive material. Sealed sources exceeding 0.001 microcuries of removable radioactive material shall be rejected.

3.11.1.4 Source radioactivity test. Each sealed source to be utilized in the AN/UDM-2 Radiac Calibrator Set must be subjected to the tests in 4.10.1.6 before it is installed in the calibrator to assure that the maximum quantity limitations for radioactive material is not exceeded.

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3.11.2 Calibrator contamination tests. Each calibrator shall be subjected to the tests of 4.10.2. The results of these tests shall not exceed 0.001 microcuries of removable radioactive material. Calibrators exceeding 0.001 microcuries of removable radioactive material shall be rejected.

3.12 Recycled, virgin and reclaimed materials. It is required that, to the maximum extent possible, recovered or reclaimed materials, in lieu of the virgin materials, shall be used without jeopardizing the intended use of the manufactured item.

3.13 Workmanship. The calibrator shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of MIL-P-11268.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own facilities, facilities of a subcontractor or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that the supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (does not include packaging) (see 4.3).

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection of packaging (see 4.14).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish nine (9) first article units of the AN/UDM-2( ) for group A, B and C testing.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in table I, and shall be performed in the order specified in table I. After completing tests of table I, each unit shall have their sources removed and the sources shall be tested in accordance with 4.10 (less 4.10.1.3 and 4.10.1.4).

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TABLE I. First article inspection.

Inspections	Requirement paragraph	Test paragraph	Order of tests Units 1 - 3	Units 4 - 6	Units 7 - 9
1. Inspections covered by subsidiary documents. 2/	3.3 3.9 3.3.1 3.13	4.4	Tests to be performed on all units.		
2. Group A inspection 2/	See Table II		Tests to be performed on all units.		
3. Group B inspection 2/	See Table III		Tests to be performed on all units.		
4. Group C inspection 3/					
Non-damaging 4/					
Altitude	3.6.2.2 3.6.3.2	4.9.5.1 4.9.5.2	3		
High temperature	3.6.3.1	4.9.1	2		
Low temperature	3.6.2.1	4.9.2	1		
Humidity	3.6.2.3	4.9.6	4		
Immersion	3.6.3.3	4.9.3		4	
Rain	3.6.3.7	4.9.12			2
Potentially damaging					
Vibration	3.6.3.6	4.9.7		1	
Shock	3.6.3.8	4.9.10		2	
Bench handling	3.6.3.4	4.9.4			1
Salt fog 1/	3.6.2.4	4.9.9	5		
Fungus 1/	3.6.3.9	4.9.11			3
Bounce	3.6.3.5	4.9.8		3	

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished, as specified in contract, before proceeding with subsequent tests. Only one equipment shall be submitted to fungus test.

2/ Inspections 1 to 3 shall be performed in the order shown before subjecting the equipments under test to any other inspection requirements.

3/ Other inspections may precede, follow, or be interspersed between the non-damaging tets.

4/ The order on non-damaging tests and the choice of which units are subjected to the non-damaging tests may be varied if convenient, except that the vibration test shall be performed on the same unit used for the bounce and shock tests.

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4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where required
Parts, materials, and processes; general	3.3
Finish	3.3.1
Marking	3.9
Workmanship	3.13

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.5.1 through 4.5.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspection in table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels and AQLs indicated in table II.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the government.

4.5.1.2 Visual and mechanical inspection. These inspections shall be performed in any order which is satisfactory to the government. The units shall be examined for the applicable defects listed in MIL-STD-252 (see 3.1).

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TABLE II. Group A inspection.

Inspection	Requirement paragraph	Inspection paragraph	AQL%	
			Major	Minor
Visual and mechanical	3.1 3.10	4.5.1.2	1.0	4.0
Functional				1/
Surface dose rate	3.4.3	4.8	1.0	
Calibration accuracy	3.4.1	4.6.1	1.0	
Compatibility	3.4.2	4.7	1.5	
Radiological				1/
Source test 2/	3.11.1	4.10.1		
Calibrator contamination 3/	3.11.2	4.10.2		

1/ No minor functional or radiological defects permitted.

2/ If the prime contractor does not manufacture the sources, this inspection shall be a requirement placed by the prime contractor on his subcontractor.

3/ This inspection shall be performed as a requirement under group A inspection (4.5.1) and also prior to packaging and packing (4.14).

4.5.2 Group B inspection. This inspection, including sampling, shall conform to table III and to procedures for special inspection levels of table I of MIL-STD-105. Group B inspection shall be performed on production lots of products that have passed group A inspection (see 4.5.2.1).

4.5.2.1 Group B sampling plans. The group B AQL shall be 4.0 percent defective and the inspection level shall be S-3.

4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in an order satisfactory to the government.

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TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test paragraph	AQL
Interchangeability	3.8	4.12	4.0%
Weight	3.5	4.13	4.0%

4.5.3 Group C inspection. This inspection shall consist of the tests specified in table IV and shall be performed on units of product that have been subjected to and passed group A and group B inspection requirements. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for group C inspection. Group C inspections shall be performed on the first production lot and at the 25 percent, 50 percent, 75 percent and 100 percent points in production. Three (3) samples of AN/UDM-2 shall be selected at random for each group of inspections and shall be performed in an order shown in table IV.

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TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
Low temperature	3.6.2.1	4.9.2
High temperature	3.6.3.1	4.9.1
Altitude 1/	3.6.3.2	4.9.5
Humidity	3.6.2.3	4.9.6
Salt fog	3.6.2.4	4.9.9
Subgroup 2		
Vibration	3.6.3.6	4.9.7
Shock 1/	3.6.3.8	4.9.10
Bounce	3.6.3.5	4.9.8
Immersion	3.6.3.3	4.9.3
Subgroup 3		
Bench handling	3.6.3.4	4.9.4
Rain	3.6.3.7	4.9.12
Fungus 1/	3.6.3.9	4.9.11

1/ These tests shall be performed only on the first production lot and on a sample selected from the lot representing approximately the 50 percent point in production. Only one equipment shall be submitted to fungus test.

4.5.3.2 Group C failures. Action required relative to group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection only.

4.6 Performance.

4.6.1 Calibration accuracy. Each calibrator, when checked with the government furnished Victoreen Model 555 Radacon II in accordance with the procedures specified herein, shall meet the requirements of 3.4.1.

4.6.1.1 Dosimeter jig assembly measurements (lower source). Interconnect the Radacon II and its accompanying remote head in accordance with the manufacturer's instruction manual supplied with the instrument; attach the probe (Victoreen Part No. 555-100-MB) to the appropriate terminals on the remote head and allow a 30 minute warmup. The lower source of the dosimeter jig assembly shall be tested in accordance with SM-A-509093 (see 3.1) and shall meet the requirements of 3.4.1.1.

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4.6.1.2 Dosimeter jig assembly measurements (upper source). Same as 4.6.1.1 except probe Victoreen Part No. 555-100-1C shall be used. The upper source of the dosimeter jig assembly shall be tested in accordance with SM-A-509093, paragraph 3.2 and shall meet the requirements of 3.4.1.2.

4.6.1.3 Dose rate jig assembly measurements. Same as 4.6.1.1. Dose rate jig assembly shall be tested in accordance with SM-A-509093 paragraph 4.0 and shall meet the requirements of 3.4.1.3.

4.7 Compatibility.

4.7.1 Dosimeter jig assembly. Calibrate all Radiacmeters IM-9, IM-93 and IM-147 supplied as Government Furnished Equipment (GFE) at mid scale using a cobalt 60 source, the calibration of which is traceable to the National Bureau of Standards. The radiacmeters shall read as follows:

IM-9( )/PD	100 mR ± 10 percent
IM-93A/UD	300 R ± 10 percent
IM-147( )/PD	25 R ± 10 percent

Any government furnished radiacmeter, whose reading exceeds the specified tolerance, shall not be used. Select 10 units of each type of radiacmeter and charge to zero. Discharge each radiacmeter in its appropriate position in the dosimeter jig assembly for the period specified for the particular radiacmeter under test. Repeat the above procedure for a total of 3 readings for each radiacmeter and calculate the average reading. The average reading shall be within the allowable tolerances specified in 3.4.2.

4.7.2 Dose rate jig assembly. Calibrate all Radiacmeters IM-174A/PD, IM-174B/PD and Radiac Sets AN/PDR-27( ) supplied as GFE using a cobalt 60 source, the calibration of which is traceable to the National Bureau of Standards. The instruments shall be calibrated at the following points:

Instrument	Calibration point	Tolerance
IM-174A/PD	100R/hr	± 10 percent
IM-174B/PD	100R/hr	± 10 percent
AN/PDR-27( )	250 mR 25 mR 2.5 mR 0.25 mR	± 20 percent

Any government furnished equipment which can not be calibrated shall not be used. Select 2 units each of Radiacmeters IM-174A, IM-174B and 2 units of Radiac Set AN/PDR-27( ) and check the calibration of each unit in the calibrator in accordance with the procedures specified. Repeat each check for a total of 3 readings per unit and the average reading per unit shall be within the allowable tolerances specified on 3.4.2.

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4.8 Surface dose rate. Using a government furnished Radiac Set AN/PDR-27 calibrated as specified in 4.7.2, the surfaces of the Radiac Calibrator AN/UDM-2 shall be monitored for conformance to the requirement of 3.4.3.

4.9 Service condition tests. Service condition tests shall be performed as detailed below. Each unit subjected to these tests shall have passed group A and group B tests and shall be resubmitted to, and pass these group A tests after completion of all service condition testing (see 3.6.1).

4.9.1 High temperature. The dosimeter jig assembly as defined by SM-D-508967 and the dose rate jig assembly as defined by SM-D-508993 shall be subjected to Procedure I, Method 501 of MIL-STD-810 with the following exceptions:

a. Temperature of step 2 shall be +155°F (68.33°C).

b. Step 3 shall be for 24 hours.

c. Step 4 shall be +125°F (51.67°C).

d. At steps 5 and 7, equipment shall meet the requirement of 3.4.2.

4.9.2 Low temperature. The dosimeter jig assembly and dose rate jig assembly, as defined in 4.9.1, shall be subjected to Procedure I, Method 502.1 of MIL-STD-810 with the following exceptions:

a. Temperature of step 2 shall be -65°F (-53.89°C) and maintained for 24 hours.

b. Temperature for step 4 shall be -25°F (-31.67°C).

c. At steps 5 and 7, equipment shall meet the requirement 3.4.2.

4.9.3 Immersion. The equipment, closed as for storage or transportation, shall be subjected to Procedure I, Method 512.1 of MIL-STD-810 (see 3.6.3.3).

4.9.4. Bench handling. The equipment, locked as for storage or transportation, shall be placed on a solid, 2 inch fir bench top. Tilt the equipment through an angle of 30° using one edge as an axis; allow to drop freely back to the horizontal. Repeat, using the remaining three edges of the same horizontal face as axes, for a total of 4 drops. Repeat the above with the equipment resting on the remaining five faces for an overall total of 24 drops. Upon completion of the test, the equipment shall meet the requirement 3.4.2 (see 3.6.3.4).

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4.9.5 Altitude.

4.9.5.1 Operating. The equipment shall be subjected to procedure I, Method 500.1 of MIL-STD-810. In step 2 test for conformance to 3.4.2 a chamber pressure of 20.6 inches of hg (10,000 ft (3048m) above sea level). Upon completion of the performance test, return the chamber to standard ambient pressure and repeat the performance test for conformance to 3.4.2 (see 3.6.2.2).

4.9.5.2 Non-operating. The equipment, closed for storage or transportation, shall be subjected to steps 2 and 4 of Procedure I, Method 500.1 of MIL-STD-810. Upon completion of step 4, the equipment shall be tested for conformance to 3.4.2 (see 3.6.3.2).

4.9.6 Humidity. The equipment shall be subjected to Procedure II, Method 507.1 of MIL-STD-810 except that:

a. The equipment shall be opened into its 2 halves during step 1.

b. Step 2 shall be deleted.

c. Conformance to 3.4.2 shall be testing during step 4.

d. Equipment shall be closed as for storage or transportation during the first 3 cycles of step 6. During cycles 4 and 5, the equipment shall be opened as in step 3.

e. Conformance to 3.4.2 shall be tested during step 8 (see 3.6.2.3).

4.9.7 Vibration. The equipment shall be subjected to Procedure X, curve AW of MIL-STD-810, Method 514.1. Upon completion of the test, the equipment shall meet the performance requirement of 3.4.2 (see 3.6.3.6).

4.9.8 Bounce. The equipment shall be subjected to Procedure X, paragraph 4.16.2, Method 514.2 of MIL-STD-810. Upon completion of the test, the equipment shall meet the performance requirement of 3.4.2 (see 3.6.3.5).

4.9.9 Salt fog. The equipment, opened as in operational use, shall be subjected to Method 509.1, Procedure 1 of MIL-STD-810. Salt concentration shall be 20 percent. Upon completion of the test the equipment shall meet the performance requirements of 3.4.2 (see 3.6.2.4).

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4.9.10 Shock. The equipment when packed for shipment shall be subjected to Procedure II, Method 516.2 of MIL-STD-810 except that the height of drop shall be 2 feet. Upon completion of the test, the equipment shall meet the performance of 3.4.2 (see 3.6.3.8).

4.9.11 Fungus. The equipment in its opened position shall be subjected to Procedure I, Method 508.1 of MIL-STD-810 (see 3.6.3.9). Upon completion of the test, the equipment shall meet the requirement of 3.4.2.

4.9.12 Rain. The equipment, closed as for storage or transportation, shall be tested as follows:

a. Dry at 150°F ±5°F (from 62.78°C min to 67.33°C max) for 48 hours.

b. Condition for 4 hours at 77°F (25°C) and 50 to 60 percent relative humidity.

c. Take pretest data. The equipment shall meet the requirement of 3.4.2.

Test equipment in accordance with Method 506.1, Procedure I of MIL-STD-810. There shall be no wind source and the rain fall throughout the test shall be 4 ± 1 inch/hr. Complete final measurements within 1 hour after the rain has been discontinued (see 3.6.3.7).

4.10 Radiological tests. These tests shall be performed by or under the direct supervision of the Radiological Protection or Radiological Safety Officer employed by the contractor in accordance with current US Nuclear Regulatory Commission regulations (see 3.11). The government reserves the right to observe and/or participate in the radiological quality assurance tests performed on each sealed source utilized in the AN/UDM-2 Radiac Calibrator Set.

4.10.1 Source tests. (see 3.11.1)

4.10.1.1 First encapsulation leak test. Fill a 200-500 ml beaker about 2/3 full of glycerine. Boil the glycerine for several minutes to drive off any dissolved air. The temperature of the glycerine shall be kept about 150°C during the remainder of the tests. A transparent beta shield (preferably plastic), having a minimum area density of 1gm/cm^2 should be placed between the individuals performing or observing the tests and the beaker, its heat sources and the shielding type source container, to protect them from the radiation. A light should be directed on the beaker at right angles to the line of view of the operator. Care shall be taken to insure that the light does not shine in the operator's eyes. Using remote handling tool(s), remove a source (at or below 30°C) from the shielded container and drop it into the hot glycerine. Closely observe the source and the glycerine above it for one minute. A large hand lens mounted near the beaker or the use of binoculars or a telescope that will focus on objects as close as two or three feet will aid in this test. Bubbles from the source or on the source indicate a leak.

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4.10.1.2 Source bloat test. After a source has passed a leak test, the source shall be tested for bloating (bulging) of the thin window. Thin window bloating (bulging) in excess of 0.030 inch shall be considered failure of the test.

4.10.1.3 Second encapsulation leak test. After the second encapsulation of the source repeat the test described in 4.10.1.1.

4.10.1.4 Second encapsulation bloat test. After a source has passed the test outlined in 4.10.1.3, repeat the test of 4.10.1.2 (see 3.11.1.2).

4.10.1.5 Source contamination test. (see 3.11.1.3)

a. Use any standard beta counting system which includes a scaler and timer and has a detector window at least one inch in diameter and not exceeding 2.0 mg/cm^2. Commercially available sponge or smooth surface filter paper wipes shall be utilized whose diameter shall not exceed the diameter of the detector window or counting planchet.

b. The beta counting system utilized in a above shall be calibrated using calibration reference standards certified by, or traceable to the National Bureau of Standards. The radioactive material incorporated into the calibration reference standards shall be of the same type as contained in the sealed sources utilized in the AN/UDM-2 Radiac Calibrator Sets (i.e., Strontium-90). A detailed procedure for the analysis of wipe test samples, including the determination of the quantity of contamination in microcuries, shall be provided the government for evaluation and approval for use.

4.10.1.6 Source radioactivity test. Each sealed source shall be tested to determine its radioactive material content by utilizing appropriate radiation measurement instrumentation to assure that the maximum quantity limitations for radioactive material contained in each sealed source type are not exceeded. The radiation measurement instrumentation shall be calibrated utilizing calibration standard sources certified by, or traceable to, the National Bureau of Standards. The radioactive material incorporated into the calibration standard sources shall be of the same type as contained in the sealed sources utilized in the AN/UDM-2 Radiac Calibrator Sets. A detailed procedure for the determination of the quantity of radioactive material contained in the sealed sources shall be provided the government for evaluation and approval for use.

4.10.2 Calibrator contamination test.

4.10.2.1 Equipment and counting procedure. The equipment and counting procedure stipulated in 4.10.1.5 above shall be followed with regards to the analysis of the wipe test samples as acquired below. Wipe test sampling shall be performed with commercially available cotton swabs in place of sponge or smooth surface filter paper wipes. To count cotton swabs, cut off all but a short stub of the handle and tape the stub to the planchet, centering the cotton on the planchet.

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4.10.2.2 Dosimeter jig assembly wipe test.

WARNING

Do not under any circumstances expose the eyes to the radiation field by peering into the access hole while the swivel cover is swung aside.

Swing aside the swivel cover and using a cotton swab, wipe around the inner surface of the access hole. Using the procedure of 4.10.2.1, check the cotton swab for contamination.

4.10.2.3 Dose rate jig assembly wipe test. Open the drawer and using a cotton swab, wipe the inside and outside surfaces of the drawer. Using the procedure of 4.10.2.1, check the cotton swab for contamination.

4.11 System safety inspection. An inspection shall be performed to verify compliance with those portions of 3.10 which can be determined visually.

4.12 Interchangeability. The mechanical dimensions shall be measured to determine conformance to the physical and functional interchangeability requirements (see 3.8). This inspection shall be conducted on piece parts and subassemblies prior to final assembly.

4.13 Weight. The equipment shall be weighed (less manuals) to determine conformance to 3.5.

4.14 Quality conformance inspection of packaging. Quality conformance inspection for packaging shall be in accordance with group A, group B and, when required (see 6.1(d)), the group C (rough handling) requirements of MIL-P-116. Inspection lots shall be in accordance with MIL-P-116.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SPI 1G00152.

6. NOTES

6.1 Intended use. The dosimeter discharge well assembly is used to check the calibration of Radiacmeters IM-9( )/PD, IM-93( )/UD and IM-147( )/PD (dosimeters). The assembly contains four each sealed sources arranged to radiate into a central cavity. The upper field is utilized in checking the operational

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reliability of Radiacmeter IM-9( )/PD while the lower field performs a similar function for Radiacmeters IM-93( ) and IM-147( )/PD. The dose rate jig assembly consists of a drawer unit and a spacer block. The drawer unit contains one each sealed source. The spacing block provides varying field intensities used to calibrate radiacmeter probes. This assembly is utilized to calibrate Radiacmeters IM-174A/PD, IM-174B/PD and Radiac Set AN/PDR-27( ).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Complete equipment or individual units to be procured.

c. Level A or B preservation and packing (see section 5).

d. When rough handling test is required.

e. Number of first articles to be submitted for approval (see 3.2).

f. Marking and shipping of samples.

g. Place of final inspection.

h. Technical literature required.

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Nomenclature. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.5 Government furnished property (loaned). Unless otherwise stated in the contract, the following government property, in the quantities specified, shall be loaned to the contractor for the purpose of performing acceptance tests on the equipment being procured:

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Radiacmeter IM-9( )/PD	25 each
Radiacmeter IM-93A/UD	25 each
Radiacmeter IM-147( )/PD	25 each
Radiacmeter IM-174A/PD	6 each
Radiacmeter IM-174B/PD	4 each
Radiac Set AN/PDR-27( )	4 each
Radiac Charger PP-1578( )	6 each
Radiac Set AN/PDR-60	4 each
Victoreen Model 555 Radacon II with instruction manual, Victoreen Model 555-100-MB, probe, Victoreen Model 555-100-1C probe, and accessories 1-5 per SM-A-509093, Timer 555-50	1 each

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